

Characterization and forecast of global influenza subtype dynamics

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Francesco Bonacina ^{1,2,6}, Pierre-Yves Boëlle¹, Vittoria Colizza ¹,
Olivier Lopez ³, Maud Thomas⁴ & Chiara Poletto ⁵ 

The subtype composition of seasonal influenza waves varies in space and time. Influenza subtypes A/H1N1, A/H3N2 and B tend to have different impacts on population groups; therefore, understanding the drivers of their cocirculation and anticipating their composition is important for epidemic preparedness. FluNet provides data on influenza specimens by subtype for more than 150 countries. However, owing to surveillance variations across countries, global analyses usually focus on subtype compositions, a kind of data difficult to treat with advanced statistical methods. We used compositional data analysis to circumvent the problem and study trajectories of annual subtype compositions of countries. Here we first examine global trends from 2000 to 2023. We identify a few seasons which stood out for the strong within-country subtype dominance due to either a new virus/clade taking over (2003/2004 season, A/H1N1pdm pandemic) or subtypes' spatial segregation (coronavirus disease 2019 pandemic). Second, we show that geographical factors, most notably international mobility, concurred in shaping countries' composition trajectories between 2010 and 2019. Trajectories clustered in two macroregions characterized by subtype alternation versus persistent mixing. Finally, we define five algorithms for forecasting the next year's composition and found that incorporating the global history of subtype composition in a Bayesian hierarchical vector autoregressive model improved predictions compared with naive methods. The joint analysis of spatiotemporal dynamics of influenza subtypes worldwide reveals a hidden structure in subtype circulation that can be used to improve predictions of the subtype composition of next year's epidemic according to place.

Since 2009, the influenza type B and the H1N1pdm and H3N2 subtypes of influenza type A have been cocirculated in the human population (hereafter collectively referred to as subtypes for brevity), with highly variable occurrences in space and time^{1,2}. The viral diversity profoundly impacts the epidemiological characteristics of epidemics: epidemics dominated by A/H3N2 strains are generally more severe as this strain is more transmissible³ and causes increased morbidity in the elderly due to the immune imprinting mechanism^{4–6}, whereas epidemics due to the A/H1N1 and B strains tend to cause more infections in the young^{6–8}. The

diversity of strains also manifests with extended epidemic periods, as, for example, the peak of A infections and of B infections are typically separated by a few weeks in the northern temperate regions^{9,10}. Anticipating the subtype composition of the coming season is therefore key to improving preparedness, for example, planning hospital capacities, awareness campaigns and vaccine distribution^{10,11}.

However, anticipating the subtype cocirculation is complicated because influenza viruses interact with one another. Evidence of viral interference has been found experimentally^{12,13}, from mathematical

A full list of affiliations appears at the end of the paper. ✉ e-mail: chiara.poletto@unipd.it

models fitted to country-level incidence data^{3,14} and from country^{10,11} and multicountry^{9,15,16} statistical analyses. In particular, past studies have shown that cross-immunity is an important ingredient of models aiming at reproducing plausible influenza dynamics^{3,14}. In other words, Influenza subtypes form a coupled ecological system that needs to be studied as a whole. A second source of complication is represented by the fact that influenza rapidly spreads globally^{17,18} and epidemics in different countries are interdependent. This makes the study of worldwide influenza circulation critical for interpreting the viral patterns observed within a focal country.

In response to these needs, the Global Influenza Surveillance and Response System (GISRS)¹ gathers and makes available through the FluNet portal the weekly number of samples by subtype and country. The quality and quantity of the data are constantly increasing, but subtyping effort is not standardized across countries yet. To allow a multicountry comparison, it is therefore preferable to focus on rescaled quantities, such as the percentages of infections by each subtype. These data allowed for showing strong patterns of alternation between A/H1N1pdm and A/H3N2 in Europe^{19,20}, the domination of A/H3N2 among all subtypes and interhemispheric synchrony in its circulation⁹. It was also used to characterize the altered subtype circulation following the emergence of the A/H1N1pdm subtype¹⁵ and the coronavirus disease 2019 (COVID-19) pandemic²¹. However, these studies did not address the spatial drivers of subtype composition nor provide anticipation. Research work attempting to forecast the next year's subtype composition in a country is so far scarce²².

Here we undertook a systematic analysis of the coupled dynamics of subtypes by analysing their relative abundances across countries and years through the compositional data analysis (CoDA) framework^{23,24}. This approach has been used in ecology and geology to analyse percentage data, taking into account the sum-to-one constraint. However, its use in infectious disease epidemiology has been notably rare²⁵. We defined a trajectory in the subtype composition space for each country. We quantified how these trajectories evolved in time and space and identified predictors of spatial structure. We then proposed an approach to leverage this structure to forecast subtype relative abundance in each country 1 year ahead with improved accuracy compared with naive estimators.

Results

Studying the relative abundances of influenza subtypes with CoDA

We studied the relative abundance of influenza subtypes for different countries/years, defined by the percentages of the form (B%, H1%, H3%). For brevity, we will use in figures and equations H1 for the A/H1N1 strains—historical A/H1N1 before 2009 and the A/H1N1pdm after—and H3 for A/H3N2. We consider weekly surveillance data reported in FluNet^{1,26} from 2000 to 2024, for up to 167 countries. We aggregated data annually, from May to April, and computed ‘compositions’²³, that is, the percentages of each of the three subtypes for each year and country (Fig. 1a) (see Methods for details). These compositions are ‘sum-to-one’ data that can be shown in a ternary plot (Fig. 1b).

The study of this type of data is complicated since the components are inherently non-independent^{23,27}, making standard statistical tools inadequate²⁴. Modern compositional analysis solves the issue by transforming the original data to log-ratios²⁸. Here we use the isometric log-ratio (i.l.r.) transformation²⁸. An alternative, the additive log-ratio transformation, was also tested for sensitivity. The i.l.r. transformation maps the three percentages (B%, H1% and H3%) onto two coordinates: the log-ratio of B% to $\sqrt{H1\% \cdot H3\%}$ and the log-ratio of H1% to H3% (refer to equation (1) in the Methods). The order of the percentages was chosen for straightforward epidemiological interpretation and does not affect the analyses. The ternary graph can then be remapped to the new coordinates (Fig. 1c), which makes it easy to represent trajectories of subtype abundance over time for a given country (Fig. 1d). In these

graphs, we also highlighted ‘dominance’ boundaries corresponding to compositions with one subtype amounting to 50% of the samples. The central region, or ‘codominance’ region, corresponds to neither subtype being dominant.

Global multiannual trends in subtype mixing

The composition trajectories for 167 countries are shown in Fig. 2a. The number of countries contributing to FluNet was initially limited but has substantially increased since then. For example, in 2000, only 31 countries contributed more than 50 samples, our threshold for inclusion, compared with 149 countries in 2023.

We first analysed global statistics on the ensemble of trajectories from one year to another. We introduced the mixing score to quantify the subtype mixing in a country/year. This is defined as the shortest distance in the log-ratio plane between the country/year composition and the boundary of the codominance region. It ranges from positive values when the composition is inside the codominance region to negative ones when it is outside, that is, one subtype is dominant. The measure has an upper bound at 0.56, corresponding to the equipartition of cases of a country/year among the three subtypes, and no lower bound. For example, strong dominance—for example, one subtype exceeding 75% frequency—yields a mixing score below, for example, −0.8, depending on how the remaining two subtypes are distributed. The distribution of countries’ mixing scores in a given year provides a global overview of country-level subtype mixing. The multiannual comparison of Fig. 2b highlights anomalous events, when the mixing score was extraordinarily low, overall. A/H3N2 was strongly dominant (that is >75%) in 35 out of 44 countries in 2003, and A/H1N1pdm was strongly dominant in 82 out of 99 countries in 2009 (Fig. 2c). One subtype among A/H1N1pdm, A/H3N2, and B was strongly dominant in 13 out of 31 countries in 2020, and in 83 out of 116 countries in 2021.

Atypical values correspond to events occurring at the global scale in those years. The strong dominance of A/H3N2 in 2003 was caused by an emerging clade for which the vaccine had limited effect^{29,30}. Similarly, the strong dominance of A/H1N1pdm in 2009 was due to the zoonotic emergence of the A/H1N1pdm subtype and the consequent pandemic^{15,31}. In both examples, a punctuated change in the virus caused a strong level of dominance, which was associated with a high level of subtype synchrony across countries, and, especially for A/H1N1pdm, more intense viral circulation globally. During 2020–2021, subtypes circulated with enhanced regional segregation³². In 2020, A/H3N2 strongly dominated in seven Southeast Asian countries, B in three tropical-belt countries, China and Afghanistan, and A/H1N1pdm circulated mainly in five African countries, Mexico and South Korea (Supplementary Fig. 3). This was due to the unprecedented contact and mobility restrictions during COVID-19^{32–34}, which caused a worldwide drop in influenza cases (only 31 countries reported at least 50 samples in 2020). Such a strong geographical pattern observed here conditioned the dynamics of influenza resurgence starting in 2022 and the return to normal levels of mixing³⁵.

In the Supplementary Fig. 8, we analysed the mixing score under additive log-ratio coordinates. We also showed that the uncertainty in the determination of the composition due to the sample size did not impact the trend described here. In addition, we show that the Shannon entropy provides a result similar to the mixing score. The advantage of the mixing score is that it provides an easy-to-read visualization of how far the composition is from codominance.

Geographical patterns in subtype cocirculation

We then looked for geographical patterns in countries’ trajectories from 2010 to 2019, the longest recent period of regular influenza circulation, between the 2009 A/H1N1pdm and the COVID-19 pandemics, and characterized by a large number of countries (86) satisfying our inclusion criteria. To identify predictors of country similarity in subtype composition, we took distances between countries’ trajectories

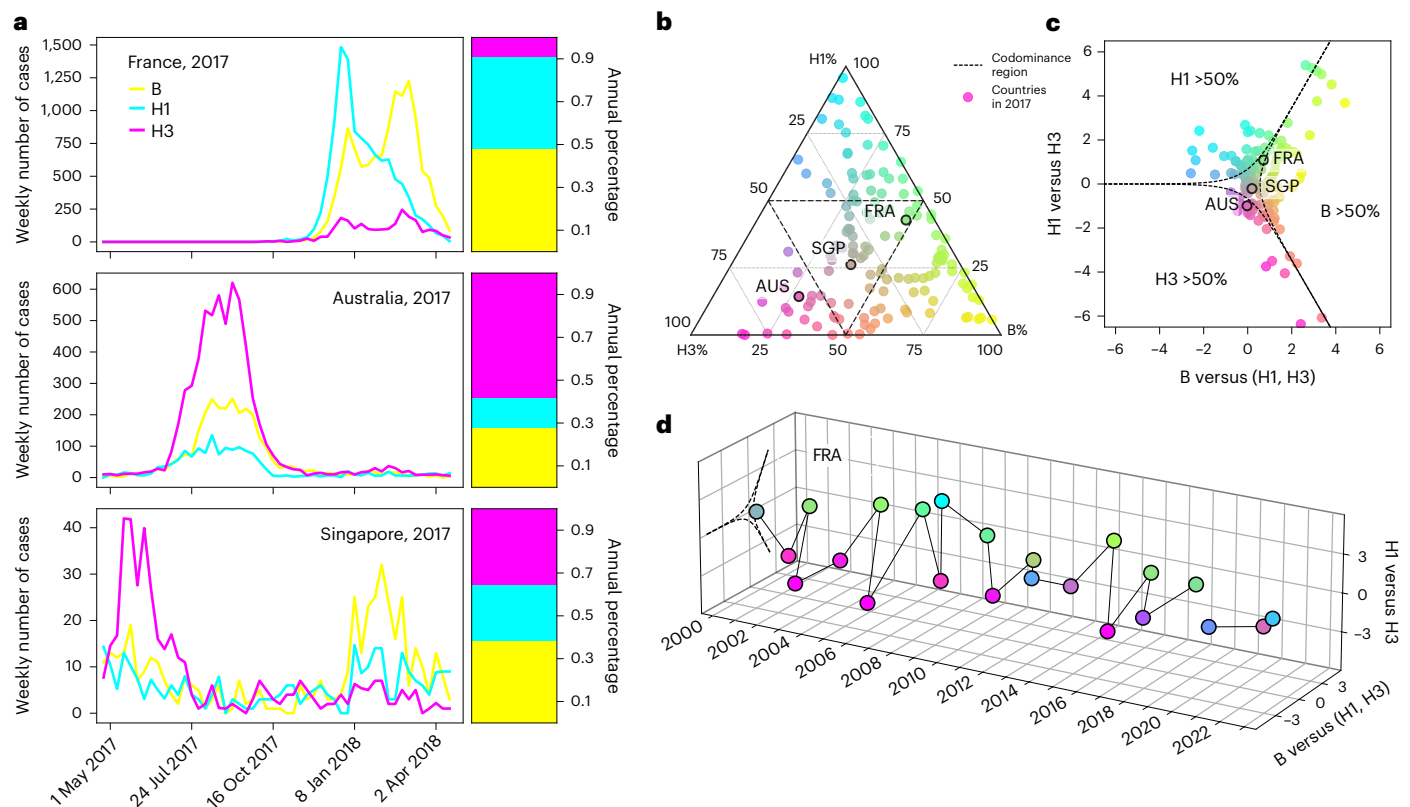


Fig. 1 | Use of CoDA to define subtype abundance trajectories. **a**, FluNet samples by subtypes for France (FRA), Australia (AUS) and Singapore (SGP). We consider the year 2017, defined as the period going from May 2017 to April 2018, as an example and compute the subtype abundances over the year. **b**, Proportions of H1 strains (A/H1N1 before 2009 and A/H1N1pdm after 2009), H3 and B represented in a ternary graph. The coloured points represent the observations for each of the 138 countries considered in 2017. The dotted lines separate the

three dominance and the (central) codominance regions. The points are colour-coded according to the proportion of B (yellow), H1 (cyan) and H3 (magenta), and the grey indicates a perfect balance of circulating strains. **c**, Relative abundances of subtypes of the same 138 points plotted in **b** after i.l.r. transformation. **d**, Trajectory of relative abundances log-ratios in France, from 2000 to 2023. The point for 2020 is missing because fewer than 50 cases were uploaded on FluNet that year. The points are coloured with the same triangular code as in **b** and **c**.

in the log-ratio plane as the response variable and applied both univariable correlations and multivariable linear regression. Predictors included country-pair differences in average temperature and humidity, inter-country air connectivity, and epidemic synchrony—classified as synchronous (same seasonal peak, same hemisphere), asynchronous (opposite hemispheres) or semisynchronous (seasonal versus year-round (YR) circulation). See Methods and ‘Geographical patterns: additional methods’ in the Supplementary Information for details. All covariates were statistically significant (Fig. 3a and Supplementary Table 2). Among the three continuous covariates, air traffic had the strongest effect. Synchrony was also very important.

To visualize the global pattern of countries’ compositions, we clustered countries with similar trajectories. Figure 3b shows the clustering in two groups. These groups comprised whole WHO influenza transmission zones (ITZs)³⁶, with only three countries allocated to the other group. Group I included 41 countries, all but one from the WHO ITZs of Europe, Northern Africa and Western Asia (Supplementary Fig. 4), whose trajectories showed strong and synchronous alternation between subtypes (Fig. 3b). The 45 countries of group II belonged to the other WHO ITZs, except for Oman and Qatar from the Western Asian ITZ (Supplementary Fig. 4). These countries displayed flatter trajectories overall, though with more variability between them than seen in group I (Fig. 3b). Group II included tropical countries characterized by cocirculation of all subtypes throughout the year, seven northern temperate countries exhibiting marked subtype alternation, yet not synchronous with group I (the USA, Canada, China, Hong Kong SAR of China, Mongolia, Republic of Korea and Japan), and countries from

South and Central America (Supplementary Fig. 4). A finer clustering in six groups split group I into two (roughly Western Asia versus Europe) and group II into the three previously described subgroups, plus Qatar (Supplementary Figs. 4 and 5), here again in good accordance with the WHO ITZs. For both the two-groups and the six-groups repartition, countries belonging to the same cluster showed more similar temperatures and/or increased connection and/or synchrony compared to the whole set of countries (Supplementary Fig. 6).

The analysis of predictors of geographical patterns was robust under changes in the log-ratio transformation and when bootstrapping the data to account for data collection uncertainties (Supplementary Fig. 9). When an alternative influenza year beginning in September was tested, results were similar except that the difference between the trajectories of asynchronous countries was smaller than in the baseline. Under a stricter criterion for inclusion of countries/years, we found that temperature and relative humidity were no longer statistically significant, whereas the influence of air travel increased (Supplementary Table 4). This may be due to the exclusion of many countries in this scenario—mainly those in the tropical belt—for which similarities in subtype composition were better explained by climate than by air connections (Supplementary Fig. 6). The country grouping was robust in all sensitivity scenarios tested (‘Alternative trajectory clusterings’ in the Supplementary Information).

Forecasting of next year’s influenza subtype composition

By examining subtype compositions as trajectories over time, we can try to predict next year’s influenza subtype composition or dominance

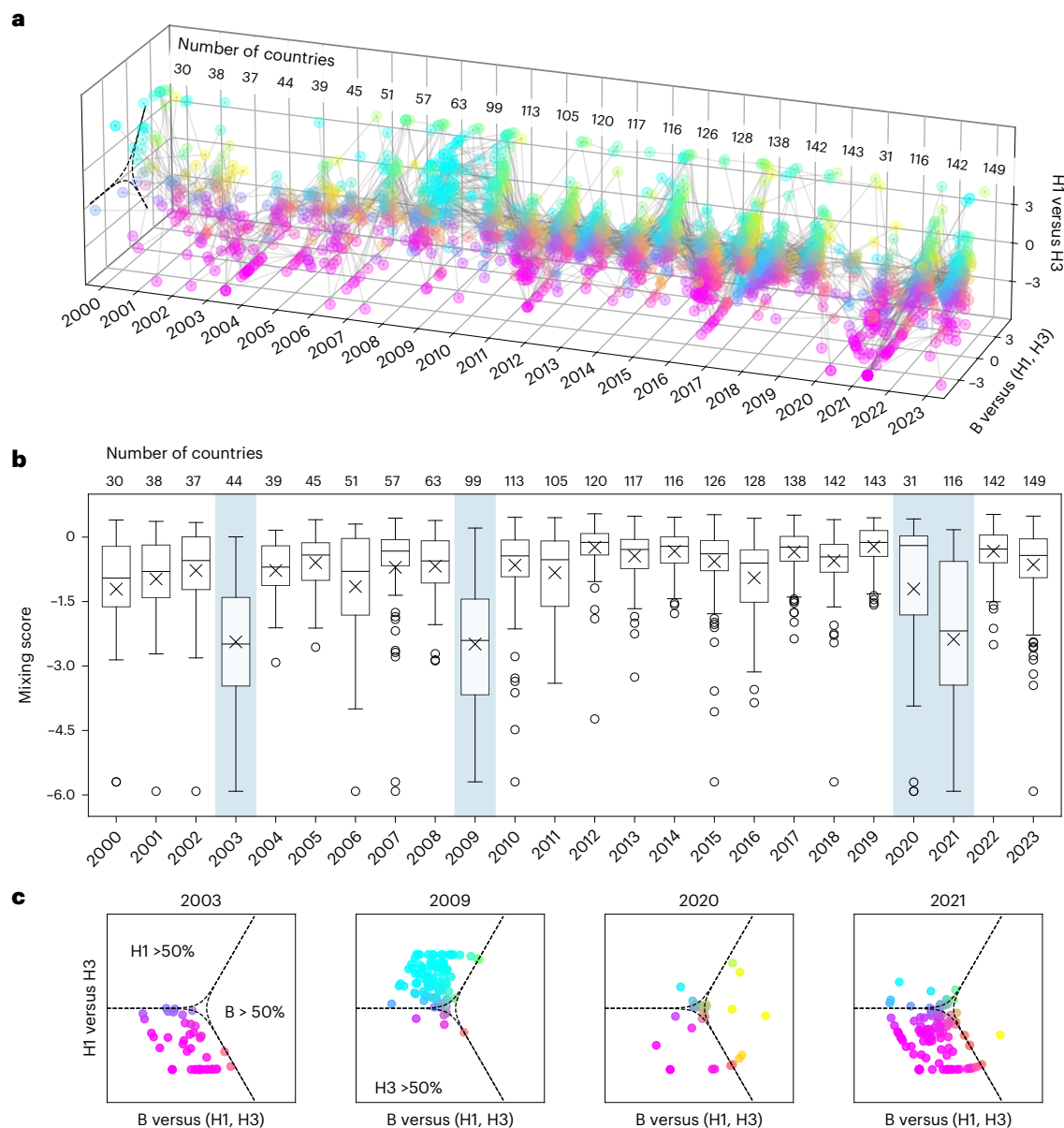


Fig. 2 | Trajectories of relative abundances of influenza subtypes and global subtype mixing. **a**, Trajectories of relative abundances of influenza subtypes. We present trajectories for 167 countries, including years from 2000 to 2023. The points in the plot represent countries/years for which the number of FluNet cases is at least 50. The number of countries included by year is shown. **b**, The degree of mixing of influenza subtypes over time. For the years 2000 to 2023, the mixing score of influenza subtypes was computed for each country, and their

distributions are depicted through the boxplots. The boxplots show a centre line for the median value, box limits for the lower and upper quartiles, whiskers for the non-outlier maxima and minima and a cross for mean value. Positive scores represent countries where each subtype is responsible for <50% of cases, whereas negative scores denote the dominance of one subtype. **c**, Influenza subtype abundances for 2003, 2009, 2020 and 2021, when subtype mixing was unusually low. The points are coloured with the same triangular code as in Fig. 1.

pattern in each country. As there is currently no accepted approach to carry out this task, we considered several possibilities, ranging from simple (naïve) models to state-of-the-art statistical models built on the CoDA framework. We focused on predicting compositions for 2017, 2018 and 2019 using the history of compositions observed worldwide since 2010. The models were as follows (see Methods for details). Model M1 ‘past frequencies’ predicts the dominance/codominance state for the coming year—between dominance of A/H1N1pdm, A/H3N2, B or codominance—as the state that occurred most often in previous years. Note that it does not predict a composition but only the dominance/codominance state. Model M2 ‘H1–H3 alternation’ propagates the composition of the past year in the same country, exchanging the percentages of A/H1N1pdm and A/H3N2, based on the evidence that these

viruses tend to alternate, whereas influenza B often codominates^{9,15,20}. Model M3 ‘average composition’ computes the average of previous years’ compositions in the same country, after log-ratio transformation. Model M4 ‘VAR’ fits a vector autoregressive (VAR) model that predicts the future composition of a country as a linear combination of its past compositions, accounting for both temporal autocorrelation and correlations between components³⁷. We used a VAR model with lag = 1. Finally, model M5 ‘HVAR’ fits a Bayesian hierarchical VAR (HVAR) model to predict each country’s composition based on both its own history and that of countries with similar trajectories^{37,38}. This approach extends the M4 ‘VAR’ model, which relies only on the history of the country under analysis and compensates for limited data over time by incorporating information on subtype compositions over

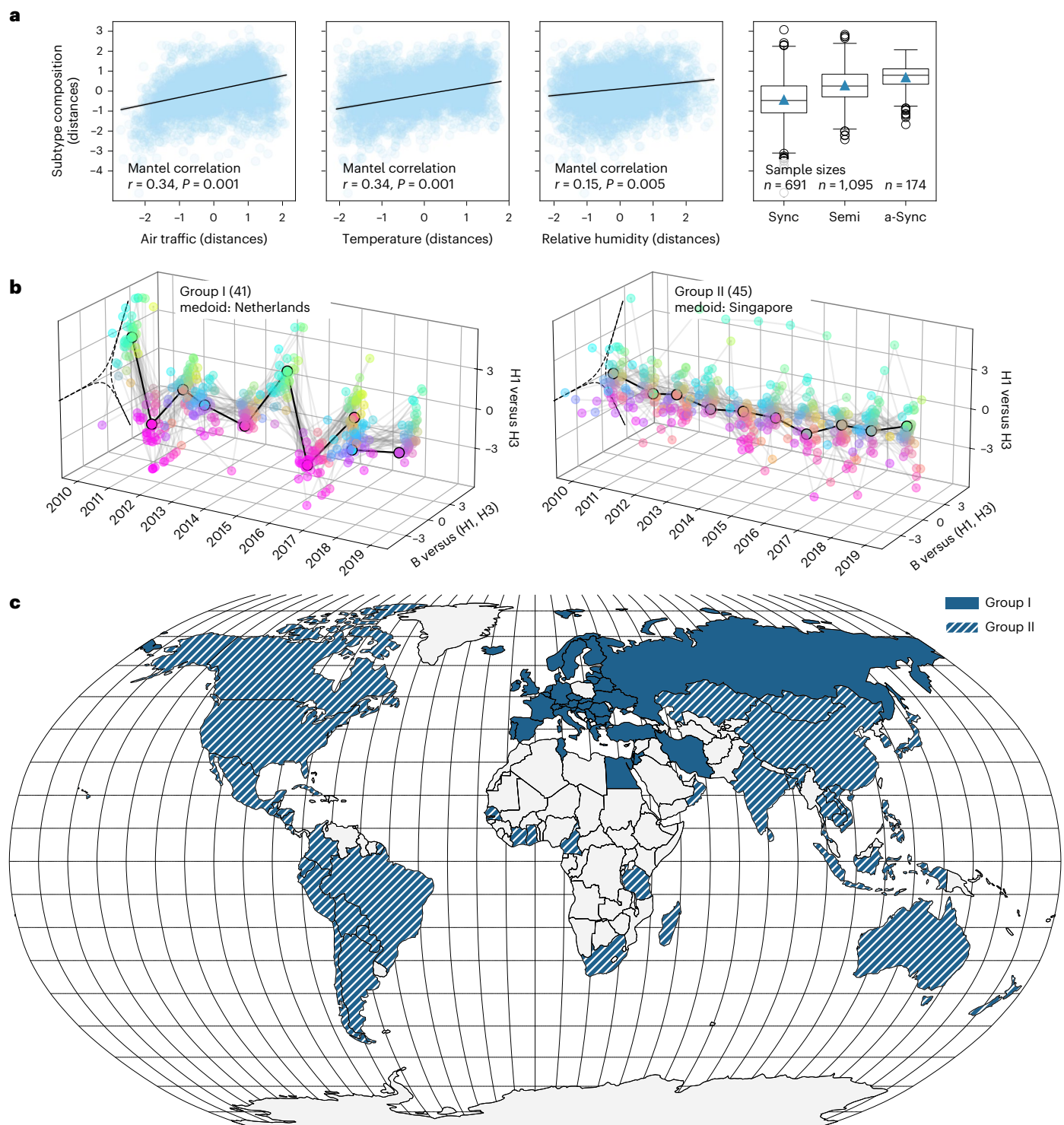


Fig. 3 | Geographical pattern of subtype trajectories. **a**, Association of country-pair subtype trajectory distances with air traffic, temperature, relative humidity and epidemic synchrony. Continuous variables were standardized, and their correlation with the response variable (r) was assessed using the two-sided Mantel's test with the Spearman method, with results reported in the legends. Linear models are also shown to highlight trends. The boxplots are used to show distributions of pairs of countries with synchronous (Sync), semisynchronous (Semi) or asynchronous (a-Sync) epidemics. The boxplots include a centre line for the median value, box limits for the lower and upper quartiles, whiskers for the non-outlier maxima and minima and a triangle for

the mean value. **b**, Hierarchical clustering of trajectories of A/H1N1pdm, A/H3N2 and B relative abundances. Trajectories for 86 countries from 2010 to 2019 are clustered in two groups—groups I and II—with 41 and 45 countries, respectively. The medoid trajectories—that is, the most central trajectories—of the two groups are highlighted in black and are reported in the legends. The codominance regions are shown with dashed lines as a reference. **c**, Geographic positioning of group I and group II countries. In the figure, H3 is used to indicate A/H3N2, and H1 is used to indicate A/H1N1pdm. Points are coloured with the same triangular code as in Fig. 1. Basemap data in **c** from Natural Earth (<https://www.naturalearthdata.com/>)⁷⁵.

Table 1 | Evaluating influenza subtype forecasting

Model	M1 'past frequencies'	M2 'H1–H3 alternation'	M3 'average composition'	M4 'VAR'	M5 'HVAR'
Predicted observable: dominance state Score: dominance state accuracy					
Group I (119)	0.235±0.003	0.227±0.003	0.218±0.003	0.193±0.003	0.277±0.003
Group II (136)	0.301±0.003	0.301±0.003	0.353±0.003	0.257±0.003	0.397±0.003
All countries (255)	0.271±0.002	0.267±0.002	0.290±0.002	0.227±0.002	0.341±0.002
Predicted observable: composition Score: energy score					
Group I (119)	–	3.813±0.010	1.811±0.008	2.194±0.007	1.426±0.005
Group II (136)	–	2.102±0.009	1.338±0.007	1.640±0.009	1.195±0.005
All countries (255)	–	2.901±0.008	1.559±0.006	1.899±0.006	1.303±0.004

Scores considered to compare the five predictive models are the dominance state accuracy for dominance state predictions and the energy score for predictions of compositions. Scores are averaged over the predicted years (2017, 2018 and 2019) and across countries within each group. Results for group I, group II and all countries appear in separate rows; brackets show the number of country/year observations—more information on the sample size is provided in the Methods. Scores are reported with uncertainties expressed as standard errors of the mean calculated over 200 bootstrap samples. For the M5 'HVAR' model, we report the result with the lag that performed the best for each group, that is, lag=2 for group I and lag=1 for group II. Methods that performed best by country groupings are shown in bold for both the dominance state accuracy and the energy score. Dominance state accuracy is positively oriented (that is, it increases as the model's performance increases), whereas the energy score is negatively oriented (that is, it decreases as the model's performance increases).

space. In practice, we first clustered the training-set trajectories as described in 'Geographical patterns in subtype cocirculation' and then fitted an HVAR model separately to each cluster, assuming that while each trajectory follows a VAR process, the VAR parameters within a cluster are not too dissimilar. We tested both lag = 1 and lag = 2. Fitted VAR coefficients are analysed in Supplementary Fig. 7.

From models M3, M4 and M5, it was also possible to compute prediction regions based on 95% probability intervals. Models M1 and M2 only allowed point predictions.

We computed forecasts for 85 countries over the 3 years (Qatar was excluded from the original 86 due to its anomalous behaviour; Methods), yielding 255 country/year predictions. Table 1, top, presents the forecast performances of the models by looking at the percentages of correctly predicted dominance states—the dominance state accuracy. Results are presented globally and for group I and group II countries. Model M1 had 27% accuracy overall, performing similarly to random guessing the dominance state out of the four possibilities, that is, dominance of A/H1N1pdm, A/H3N2, B or codominance. Accuracy was lower for group I (24%) than for group II (30%), consistent with the difficulty in predicting the marked subtype alternation of group I countries. Other models (M2, M3 and M4) had performances close to model M1. Estimates improved with M5 'HVAR', with a global accuracy of 34%, improving in both group I (28%) and group II (40%).

Models M2, M3, M4 and M5, which generate continuous predictions of subtype compositions, were also compared using probabilistic forecasting metrics commonly adopted in epidemiology³⁹, where both calibration (closeness to the target) and sharpness (prediction uncertainty) are evaluated. In Table 1, bottom panel, we show the values of the energy score⁴⁰, where smaller values correspond to better predictions. We found again that the M5 'HVAR' model performed better than other models. Predictions for France (group I) and Australia (group II) using this model are illustrated in Fig. 4 as examples.

Finally, we addressed simpler prediction tasks still relevant for public health. Precisely, for each subtype, we asked whether it would be (1) dominant (>50% of cases) or not and (2) negligible (<10% of cases) or not. For model M1 'past frequencies', the probability of negligibility of a subtype corresponded to the proportion of times the subtype was negligible in the previous years. To evaluate the performance of the five models in carrying out the two tasks, we used the area under the receiver operating characteristic curve (AUROC). Results are provided in Table 2.

The M5 'HVAR' model was the best-performing model in predicting the negligibility of influenza type B, the dominance of subtype A/H1N1pdm, and both the dominance and negligibility of subtype

A/H3N2. In particular, the M5 'HVAR' provided in general the best performance and the greatest improvement when applied to the trajectories of group I countries, in contrast with the forecasts of dominance and composition being less good for this group, as discussed above. For these countries, the AUROC score for the M5 'HVAR' model reached 0.91 in predicting B negligibility, compared with scores of 0.35–0.61 obtained with other models. Similarly, it reached 0.76 (0.80) when predicting the dominance (negligibility) of A/H3N2, compared with 0.43–0.64 (0.38–0.62) for the other models. M5 'HVAR' was not always the best-performing model, notably, it was not always able to improve the predictions of the M1 'past frequencies' and M3 'average composition' models in predicting the dominance of B and the negligibility of A/H1N1pdm. For the dominance of A/H1N1pdm for group I countries, no model performed appreciably better than a random guess. Predictions for all countries and years, obtained with the five methods, are reported in the Supplementary Data 1.

Results on model ranking were overall consistent when evaluating models with alternative metrics and when performing predictions under alternative scenarios, that is, using alternative log-ratio coordinates, a different definition of the influenza year, using the HVAR based on the six-group repartition instead of the two-group, restricting to countries/years with at least 500 cases and restricting to the 50% of countries that were best classified into the two clusters (Supplementary Tables 1, 3 and 5–7). The forecast for the 50% best-classified countries outperformed the baseline for seven out of eight observables. Notably, AUROC improved from 0.62 to 0.82 in predicting the dominance of B with M5 'HVAR', the best-performing model in this scenario. Stronger similarity with other countries in the same cluster may have improved the quality of the spatial information used for the fit in the HVAR model.

Discussion

Viral composition plays a key role in shaping influenza dynamics and the population burden of epidemics. A better anticipation of seasonal epidemics requires an improved ability to monitor, quantify and model subtype composition. This is increasingly within our reach thanks to the rapid increase of virological data on a global scale. Still, novel quantitative frameworks are needed to better visualize and analyse these data to extract meaningful information for response planning. Here we have shown that CoDA enables defining subtype composition trajectories, thus opening the door to state-of-the-art statistical tools for analysing their spatiotemporal properties and forecasting their future evolution.

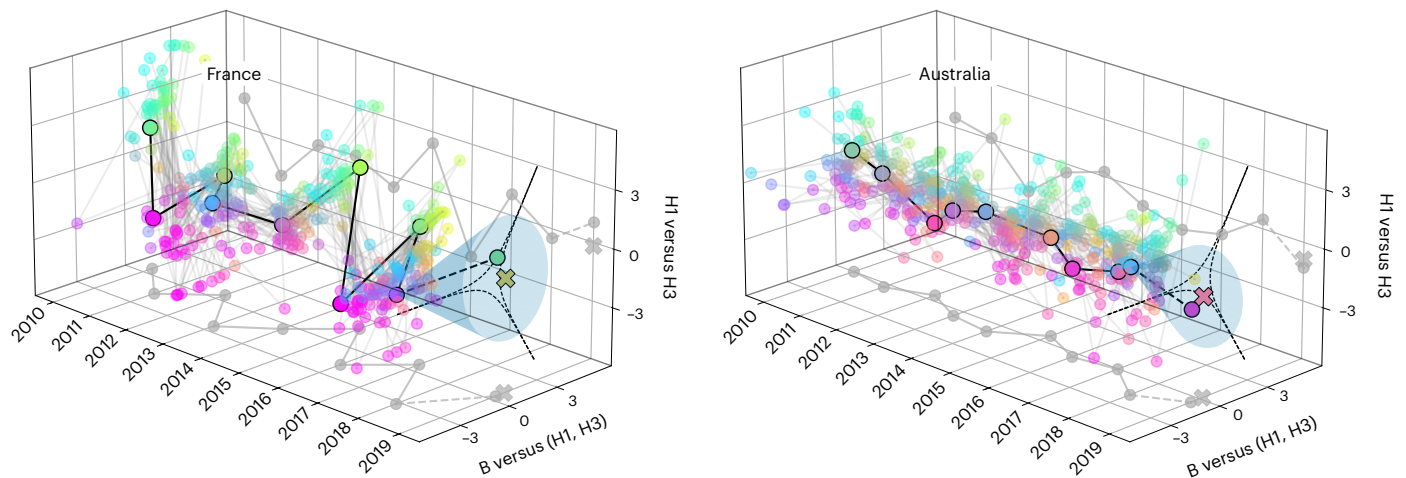


Fig. 4 | Prediction of relative abundances of influenza subtypes for France and Australia in 2019. Predictions are computed using the M5 ‘HVAR’ model of lag 2 and 1 (Methods), respectively. For each country, the observed trajectory from 2010 to 2018 is represented with a black solid line, while the dashed segment links the points corresponding to 2019, to be predicted, to the rest of the time series. Projections of the trajectories are added on the vertical and horizontal planes as a guide to the eyes. The thinner grey lines correspond to the trajectories of the

other countries within the respective group—group I for France and group II for Australia—that were used to train the model. The crosses depict the predictions and the shadow areas of the ellipses are associated with the 95% confidence intervals. The points are coloured with the same triangular code as in Fig. 1. The codominance regions are shown with dashed lines as a reference. The energy scores for the 2019 predictions for France and Australia are 0.90 and 0.66, respectively.

We have analysed subtype composition at the global level, considering trajectories of annual compositions by country. This has revealed meaningful spatiotemporal patterns. We have characterized the evolution in time of the ensemble of trajectories by focusing on the extent of subtype mixing through the mixing score. We have highlighted years of strong dominance of a given subtype within each country—the strong dominance of A/H3N2 and A/H1N1pdm in almost all countries globally, respectively occurring in 2003 and 2009, and the spatial segregation of subtypes occurring concomitantly to the COVID-19 pandemic. Previous studies on seasonal influenza disruption have primarily focused on changes in peak timing and wave severity^{15,41}. Strong subtype dominance can have marked effects on the age profile of cases. For instance, mortality shifted towards young individuals during the 2009 A/H1N1pdm pandemic⁴² probably due to residual immunity in older cohorts from early-life exposure to A/H1N1 viruses circulating before 1950 (ref. 43). In addition, a sudden reduction in viral diversity can have long-term consequences for influenza ecology, as exemplified by the disappearance of B/Yamagata since the COVID-19 pandemic^{34,44}. The mixing score introduced here provides a concise metric for monitoring trends and comparing different locations, supporting surveillance and epidemiological interpretation. In this study, we focused on the impact of extreme events on subtype mixing. Beyond this, the score could be used as a response variable to test various predictors of dominance versus codominance, such as vaccine efficacy, vaccination coverage, population’s immune memory from past epidemics and cross-immunity effects. This application would require compiling comprehensive, multicountry data on vaccine efficacy and coverage.

During a period free from disruptive events, we investigated the geographical drivers of subtype alternation. We found that pairs of countries with stronger air-travel connectivity, same epidemic periods, similar temperatures and, to a lesser extent, relative humidity, exhibited more similar composition trajectories. While cross-immunity and waning immunity are known to be key drivers of subtype alternation^{3,9–11,14–16,45}, we found that geographical factors also played a substantial role. Most notably, we identified an important role of air travel in homogenizing subtype composition across well-connected countries, especially in the north, leading to phase-locking of subtype alternation. International mobility was previously found

to shape global phylogenies of the individual subtypes^{17,18}. Yet, no study addressed the role of this ingredient on global subtypes’ coexistence so far. The emerging spatial structure was synthesized by the clustering analysis, which led to noteworthy results. Group I—that is, Europe, North Africa and West Asia—displayed a similar alternation of subtypes, clearly distinguishable from the rest of the world. While such a strong similarity was noted before¹⁹, here we show that it was due to the strong mobility coupling within the region compared with the rest of the world, together with the consistent climatic conditions and synchrony, confirming previous hypotheses¹⁹. Furthermore, North America belonged to group II and clustered with Eastern Asia in the finer clustering. Following H1N1pdm emergence, North America had a different alternation of A subtypes than Europe^{15,20}. This was previously linked to the higher vaccination coverage in North America, which mitigated the circulation of A/H3N2, preventing it from dominating in the 2011/2012 season¹⁵. Although this factor may have played a role, we found that North America’s trajectories may have entered into phase with Eastern Asia’s ones due to the strong connectivity between the two regions. Finally, Australia and New Zealand had a different subtype alternation than northern hemisphere (NH) countries a few months later. Southern hemisphere (SH) countries are often regarded as sentinels for the approaching influenza season in northern countries^{46–48}. Still, the epidemiological evidence supporting this practice is contrasting^{22,49–51}. For instance, the correlation between spatiotemporal indicators of Australia’s wave and the waves in the UK, the USA and Canada was limited^{50,51}.

Grouping regions with similar influenza patterns is essential for surveillance. To this end, WHO defined 18 Influenza Transmission Zones³⁶. Although previous studies evaluated the validity of the partition^{19,52,53}, only a few used subtype as a basis for their partition¹⁹. Here, restricting the focus to subtype proportions, we found a subdivision broadly aligned with WHO ITZs, but coarser. Consistent with air travel being an important driving force of influenza composition, some clusters included noncontiguous yet strongly connected regions, for example, North America and East Asia. This suggests the importance of considering mobility, together with climate and spatial proximity, when defining surveillance zones. Repeating this analysis after COVID-19 could reveal whether the patterns still hold and help to identify reliable priors for subtype circulation across regions.

Table 2 | Forecast evaluation for predicting the dominance (yes/no) and negligibility (yes/no) of subtypes

Model	M1 'past frequencies'	M2 'H1–H3 alternation'	M3 'average composition'	M4 'VAR'	M5 'HVAR'
Predicted observable: B will be dominant (>50%) in the next season: yes/no? Score: AUROC					
Group I (119)	0.452±0.004	–	0.495±0.004	0.425±0.005	0.641±0.004
Group II (136)	0.715±0.006	–	0.693±0.005	0.463±0.008	0.368±0.009
All countries (255)	0.551±0.003	–	0.624±0.003	0.496±0.003	0.618±0.004
Predicted observable: H1 will be dominant (>50%) in the next season: yes/no? Score: AUROC					
Group I (119)	0.489±0.004	–	0.414±0.004	0.327±0.004	0.480±0.004
Group II (136)	0.440±0.004	–	0.505±0.004	0.527±0.004	0.584±0.004
All countries (255)	0.481±0.003	–	0.485±0.003	0.439±0.003	0.529±0.003
Predicted observable: H3 will be dominant (>50%) in the next season: yes/no? Score: AUROC					
Group I (119)	0.433±0.004	–	0.478±0.005	0.640±0.004	0.764±0.003
Group II (136)	0.552±0.004	–	0.576±0.004	0.511±0.004	0.618±0.004
All countries (255)	0.506±0.003	–	0.539±0.003	0.578±0.003	0.702±0.003
Predicted observable: B will be negligible (<10%) in the next season: yes/no? Score: AUROC					
Group I (119)	0.412±0.004	–	0.349±0.003	0.610±0.004	0.905±0.002
Group II (136)	0.521±0.006	–	0.507±0.005	0.527±0.004	0.648±0.004
All countries (255)	0.561±0.003	–	0.506±0.003	0.603±0.003	0.815±0.002
Predicted observable: H1 will be negligible (<10%) in the next season: yes/no? Score: AUROC					
Group I (119)	0.699±0.005	–	0.645±0.006	0.281±0.006	0.253±0.006
Group II (136)	0.588±0.004	–	0.580±0.004	0.548±0.004	0.605±0.006
All countries (255)	0.596±0.003	–	0.567±0.004	0.423±0.004	0.460±0.004
Predicted observable: H3 will be negligible (<10%) in the next season: yes/no? Score: AUROC					
Group I (119)	0.375±0.004	–	0.447±0.005	0.715±0.004	0.796±0.003
Group II (136)	0.620±0.004	–	0.553±0.005	0.654±0.006	0.655±0.005
All countries (255)	0.565±0.003	–	0.587±0.003	0.712±0.003	0.758±0.003

Results are summarized in six panels: the three top ones focus on dominance and the three bottom ones on negligibility. As a score of prediction, we used the AUROC. Scores are averaged over the predicted years (2017, 2018 and 2019) and across countries within each group. Results for group I, group II and all countries appear in separate rows; the brackets show the number of country/year observations—more information on the sample size is provided in the Methods. Scores are reported with uncertainties expressed as standard errors of the mean calculated over 200 bootstrap samples. The probability of dominance and negligibility of subtypes cannot be calculated with the M2 'H1–H3 alternation' model; therefore, the AUROC cannot be computed for this model. Scores for the methods performing best by country grouping are shown in bold.

The CoDa transformation enabled the use of state-of-the-art forecasting algorithms such as 'VAR' and hierarchical VAR to predict the next year's subtype composition. Trained on 2010–2019 data to avoid periods of disrupted circulation, 'VAR' was limited by the short time series and performed no better than naive estimates. Still, 'HVAR' improved performances by leveraging the geographical structure identified by the clustering. Hierarchical VAR is directly applicable to time series expressed in log-ratio coordinates, whereas a similar analysis performed directly on percentage data would require defining and developing ad hoc algorithms. Interestingly, despite group I countries being characterized by a marked A subtype alternation from one year to another, the M2 'H1–H3 alternation' model did not perform well, probably due to its inability to capture the trend of B proportion. Besides forecasting subtype proportions and the dominant subtype, we attempted to forecast binary outcomes, for example, whether one specific subtype would be dominant (yes/no) or negligible (yes/no). Predictions were reasonably accurate for B negligibility (AUROC of 0.82 averaged over all countries) and A/H3N2 negligibility (0.76) and dominance (0.70), especially for group I countries. Predicting the dominance of A/H3N2 is important for preparedness since this is often associated with a high epidemic burden.

Epidemic forecasting is a central problem in infectious disease epidemiology. Consortia of modellers are dedicating a great effort to projecting seasonal influenza epidemics in the short and long term. In the context of these initiatives, a wide range of mechanistic and statistical models are being combined in ensemble modelling to forecast influenza incidence 1–4 weeks ahead or to identify possible evolution scenarios months ahead^{54–58}. This provides critical information for response planning. A different question regards the evolution of the influenza viruses. Several studies combine genetic and antigenic analyses to predict changes in the frequency of circulating clades for each subtype^{59,60}. The identification of the clade more likely to be prevalent in the future informs the choice of vaccine composition, which needs to be done 6 months in advance. Here we focused on a different task as we attempted to forecast subtype abundance by country one year ahead. We have framed the problem by introducing a range of models from the more naive to the more sophisticated. The subtype composition affects the expected epidemic size, the peak timing and the age distribution of cases^{3,6,9,11}. Predictions of next year's subtype composition could support scenario modelling. The dominant subtype has been identified by the Flu Scenario Modeling Hub⁵⁸ as key information for projecting hospitalizations

and deaths. This is essential for planning epidemic response ahead of the season, including vaccine and hospital resource allocation. In the short term, forecasting efforts pay special attention to the epidemic's start, peak, severity and duration, whose knowledge is critical for epidemic management. Knowing the dominant subtype at the start of the season would inform and complement forecasts of these features. For instance, it would allow the use of statistical forecasting algorithms trained only on past seasons with similar subtype frequencies, leading to higher prediction accuracy. It would also provide information on age-specific severity, enabling healthcare providers to be alerted and recommendations to be better targeted.

The HVAR model introduced here can be improved in the future. Restricting to the 50% best-classified countries improved the forecast for many observables. Although identifying more predictable countries is informative for preparedness⁶¹, this also suggests that incorporating information on other countries' trajectories at different degrees according to the similarity could increase the quality of predictions. The HVAR model leveraged the clustering to coarsely account for the spatiotemporal pattern of subtype composition. More sophisticated models could explicitly account for the drivers of such a pattern. For instance, instead of relying on the geographical cluster, we could explicitly add the dependence on all other countries' compositions, weighted by air travel distance. We could add to the VAR model terms for average temperature, humidity and epidemic period. Antigenic drift and vaccination could be accounted for by incorporating their effect on dominance⁴⁵, leveraging recent works to forecast influenza evolution⁶². Finally, new models could be designed to capture causal relationships between epidemics in different countries/years, for example, through Bayesian networks⁶³.

CoDA is under-exploited in epidemiology. Epidemiological research has traditionally focused on case numbers, which provide essential information for risk evaluation and public health assessment. However, with the growing availability of virological data—especially post-COVID-19—there is an increasing need to analyse frequencies, which are key to understanding ecological interactions between pathogen strains and are more amenable to comparison across surveillance systems. Analyses similar to the ones presented here could be applied to influenza subtype abundances at different spatiotemporal scales. For instance, weekly data would enable investigating within-season relative dynamics between A and B types^{11,64}, addressing the shift between the two types' waves and investigating anomalous years⁶⁴. Looking at data with higher temporal resolution would also enable investigating the potential improvement of the composition forecast as the season approaches and unfolds¹¹. This would require forecasting methodologies that are different from the ones introduced here. Within-country spatial coupling between compositions could also be investigated. More broadly, there is a variety of circulating viruses with distinct strains interfering with one another—SARS-CoV-2 variants, RSV types, HPV genotypes and Dengue serotypes, to name a few. Percentage data are also used to monitor the cocirculation of bacterial strains. This is the case when studying the dynamics of strains with different antibiotic resistance profiles of *Staphylococcus aureus* or *Streptococcus pneumoniae*^{65,66}.

Our study is affected by limitations. Statistical forecasting models can be applied to periods of stable strain circulation, as was the case for 2010–2019. Following the COVID-19 pandemic, the global circulation of influenza viruses was highly perturbed, making any prediction more difficult. Reliable composition projections depend on FluNet data accurately representing countries' viral compositions. Comparing sentinel and non-sentinel data showed a similarity between the two sources overall, although this varied across countries/years ('Initial data processing' in the Supplementary Information). Sentinel data are collected via a standardized protocol and are thus better interpretable. However, they contribute only to a small

fraction of FluNet data at present. We have aggregated B/Victoria and B/Yamagata because lineage information for B samples was scarce; for example, only four countries, among the ones satisfying the inclusion criteria, had at least 50% of B samples or at least 100 B samples, regardless of the percentage over the total, further subtyped in lineages for each year in the period 2010–2019. Still, the CoDA framework can be directly extended to the four subtype system whenever these data are available, thus enabling the investigation of the competitive dynamics between B lineages and their differential cross-immunity with A subtypes.

Methods

FluNet data

FluNet publishes weekly global data on influenza testing by country, reporting the number of samples tested and their results—positive or negative—categorized by influenza A subtypes, B lineages and surveillance type (sentinel, non-sentinel or undefined) ('Initial data processing' in the Supplementary Information)^{1,26}. Here we use for brevity H1 for A/H1N1 and A/H1N1pdm and H3 for A/H3N2. For each country, we determine influenza B infections by summing B/Yamagata, B/Victoria and unspecified B samples. We grouped the prepandemic A/H1N1 and A/H1N1pdm. The untyped influenza A samples were distributed between H1 and H3 according to the respective proportions of these subtypes among the subtyped A samples that week. If no A samples were subtyped that week, we looked at the proportions of H1 and H3 in the 5 weeks centred around the week or in the year. We aggregated influenza B, H1 and H3 samples over 1 year and calculated the respective percentages to define the relative abundance of the three influenza strains. Log-ratio transformations are not defined when any component equals zero. We thus replaced zeros with small values through a Geometric Bayesian-multiplicative treatment ('Initial data processing' in the Supplementary Information). The year considered in the analyses starts from week 17. As both the 2009 influenza and the COVID-19 pandemic began in spring, we set the year's start in the spring to align southern temperate countries with northern temperate countries in terms of when the two pandemics affected the respective influenza seasons. The exact week was chosen by looking at global incidence profiles and minimizing the risk of splitting an epidemic across years (Supplementary Fig. 1). After verifying that subtypes' proportions were similar in sentinel and non-sentinel data (Supplementary Fig. 2), we aggregated the data regardless of the surveillance type. In all analyses, we discarded countries/years with fewer than 50 positive samples. In the sensitivity analysis, we tested alternative datasets using a different cut-off week for the definition of the year (week 37 instead of 17) and a threshold of 500 cases instead of 50, for country inclusion. We also quantified the impact of the uncertainty on the subtype composition due to the sample size.

i.l.r. transformation

We provide here the exact expression of the i.l.r. transformation used to map compositions in unbounded euclidean coordinates. The three-component vectors of country/year subtype abundances (B%, H1% and H3%) were transformed into two-dimensional vectors (u , v) by using the following equations

$$\begin{cases} u = \sqrt{\frac{2}{3}} \ln \frac{B\%}{\sqrt{H1\% \cdot H3\%}} \\ v = \sqrt{\frac{1}{2}} \ln \frac{H1\%}{H3\%} \end{cases} \quad (1)$$

Definition of the mixing score

The mixing score is defined as the distance between the point in the log-ratio plane (u , v), representing the subtype composition, and the boundary of the codominance region, taken with a positive sign when the point is within that region and with a negative sign otherwise. The boundary of the codominance region in the simplex is identified by

the points (B%, H1% and H3%) for which one component corresponds to exactly 50%. As an example, the points such that H1% = 50%, are mapped by the i.l.r. transformation into the coordinates $(u, v(u))$ such

$$\text{that } v(u) = \sqrt{2}(\ln(e^{u\sqrt{3/2}} + \sqrt{e^{u\sqrt{6}} + 4}) - \ln 2).$$

Geographical predictors of annual trajectories of subtype compositions

We considered the 86 countries satisfying our inclusion criteria for the period 2010–2019. We defined the distances between pairs of country trajectories as the average Euclidean distance of the corresponding points in the (u, v) plane. We gathered the number of air-travel passengers between countries, countries' average temperatures, relative humidities and influenza seasons, among the NH, the SH and YR. Then, we computed three continuous covariate matrices—the differences in temperatures and relative humidity and the air travel distance between country pairs⁶⁷—and one categorical variable matrix describing epidemic synchrony, where country pairs were synchronous, that is, NH–NH, SH–SH or YR–YR, asynchronous (that is, NH–SH) or semisynchronous (that is, SH–YR or NR–YR). We then tested the association between trajectory distance and covariates using both univariable and multivariable analysis. In the 'Geographical patterns: additional methods' of the Supplementary Information, we detail the data sources for covariates, the definition of covariates, the regression analyses and the sensitivity analyses performed.

Clustering of trajectories

We applied Ward's linkage hierarchical clustering⁶⁸ to cluster countries with similar trajectories. Hierarchical clustering provides a hierarchy of nested partitions. We used the Silhouette coefficients⁶⁹ to choose the optimal number of clusters, that is, the level of hierarchy. The optimal partition grouped countries into two groups. Robustness checks and sensitivity analyses (including alternative clustering algorithms) are described in the 'Robustness checks and sensitivity analyses' of the Supplementary Information.

Forecasting of trajectories

We tested five forecasting algorithms to predict next year's subtype composition in each country. We attempted to predict observables of different nature: (1) the multilabel categorical variable corresponding to the dominance state (dominance of B, H1, H3 or codominance), (2) the subtype composition and (3) the binary variables corresponding to the dominance (yes or no) and the negligibility (yes or no) of each subtype. Predictions of the three types of observable were associated with a quantification of the uncertainty in the prediction and a metric for prediction evaluation. However, not all observables, quantifications of uncertainty and prediction evaluations were computable with all prediction methods. The five forecasting algorithms and the evaluation metrics used in the main paper were introduced in the 'Results'. We provide additional details hereafter. A complete summary of what can or cannot be computed across predicted observables and prediction methods is reported in the Supplementary Table 1.

Algorithms for trajectory forecasting. We studied annual bivariate trajectories of compositions of the form $(y_1, \dots, y_T)$, such that $y_t = (u, v)'_t$ is i.l.r. $((B\%, H1\%, H3\%))'_t$, with $t = 1, \dots, T$ and the prime symbol indicating transposed. We forecasted y_{T+1} , that is the composition of 2017, 2018 or 2019, each time using the preceding period as the training set, that is $(y_1, \dots, y_T)$ with y_1 the composition in 2010. Below, we present the five forecasting methods.

M1 'past frequencies'. The dominance state was predicted to be the most frequently observed state during the preceding years if that state was unique. Otherwise, it was randomly chosen with uniform probability among the dominant states observed most often.

M2 'H1–H3 alternation'. The prediction for the y_{T+1} composition was $\hat{y}_{T+1} = (u, -v)'_T$, that is, in percentage form, $(B\%_{T+1} = B\%_T, H1\%_{T+1} = H3\%_T, H3\%_{T+1} = H1\%_T)$.

M3 'average composition'. The predicted composition was $\hat{y}_{T+1} = \frac{1}{T} \sum_{t=1}^T y_t$, and the prediction's confidence interval was given by the covariance matrix $\hat{\Sigma} = \frac{1}{T(T-1)} \sum_{t=1}^T (y_t - \hat{y}_{T+1})'(y_t - \hat{y}_{T+1})$.

M4 'VAR'. Assuming that the trajectory has been generated by a VAR process of lag p , then composition t can be written as a linear function of the previous p compositions: $y_t = v + A^{(1)}y_{t-1} + \dots + A^{(p)}y_{t-p} + \epsilon_t$, where v is the vector intercept, $A^{(i)}$ are 2×2 coefficient matrices and ϵ is the Gaussian noise. This can be rewritten in matrix form as $Y = BZ + U$ (ref. 37), where

$$\begin{aligned} Y &= (y_{p+1}, \dots, y_T), \\ B &= (v, A^{(1)}, \dots, A^{(p)}), \\ Z_t &= \begin{pmatrix} 1 \\ y_p \\ \vdots \\ y_1 \end{pmatrix} \\ Z &= (Z_p, \dots, Z_{T-1}), \\ U &= (\epsilon_{p+1}, \dots, \epsilon_T). \end{aligned}$$

The resulting least squares estimator is $\hat{B} = YZ'(ZZ')^{-1}$. It follows that the prediction of the composition $T+1$ is $\hat{y}_{T+1} = BZ_T$, and the prediction's confidence interval is estimated via the empirical covariance matrix corrected for short time series $y_{t,c} = v_c + A_c^{(1)}y_{t-1,c} + \dots + A_c^{(p)}y_{t-p,c} + \epsilon_{t,c}$.

We only considered VAR processes with $p = 1$, since for the shortest trajectories (prediction of 2017, with training set between 2010 and 2016), when $p > 1$, the estimator of the covariance matrix is not defined.

M5 'HVAR'. For all three forecasted years, the trajectories of the training set were clustered as described above. This yielded two groups that were almost identical to group I and group II obtained for the whole period 2010–2019 and shown in Fig. 3. Specifically, the clustering was stable and the same as in Fig. 3 for 82 out of 86 countries while it was not stable for Czechia, Greece and Russia. Qatar clustered alone for all three forecasted years and, due to its outlier behaviour, was excluded from the forecast performance evaluation. For each cluster, we assumed that each trajectory followed a VAR process, such that the process for country c is $y_{t,c} = v_c + A_c^{(1)}y_{t-1,c} + \dots + A_c^{(p)}y_{t-p,c} + \epsilon_{t,c}$. Similar to the previous model, the coefficients can be encoded in the matrix $B_c = (v_c, A_c^{(1)}, \dots, A_c^{(p)})$ and the Gaussian noise in the matrix U_c . Then, the hierarchical structure is imposed by assuming that the VAR processes for the trajectories in the group are similar. Specifically, we define $B_c = W + V_c$, with W being the matrix of coefficients encoding the average behaviour of the group, which is the same for all the trajectories in the group, and V_c being the coefficient matrix for the single trajectory adjustment. Moreover, we assume that elements in W , V_c and U_c are independent random variables, sampled from distributions parametrized by some latent variables. All model equations are shown in 'Forecast: additional methods' of the Supplementary Information. The model coefficients were optimized by maximum likelihood estimation using Monte Carlo methods. In particular, since for each coefficient it was possible to write the model's likelihood conditional on the other coefficients, efficient estimation was performed using a Gibbs sampler. For the conditional likelihood distributions and the code to implement the Gibbs sampler, we followed³⁸, modifying their model to introduce the intercept terms. From the Monte Carlo chains, for each country, we sampled the distribution of predictions $\hat{y}_{T+1}$. Then, from those samples, we computed

the mean prediction and the covariance matrix for the confidence intervals. We tested HVAR models with $p = 1$ and $p = 2$.

Metrics for evaluating the goodness of predictions. To enable model comparison, the evaluation scores of different countries/years were averaged to obtain a single value for the model performance. We associated uncertainty with the forecast evaluation by computing a confidence interval with bootstrap. This corresponded to the standard error of the mean of the evaluation metrics over 200 bootstrap samples of the set of predictions. The values reported in Table 1 correspond to the averages of the dominance state accuracy and energy score values for different countries/years, whereas the values in Table 2 are averages of the country/year AUROC scores.

To compare dominance state predictions (Table 1), we defined the dominance state accuracy as the percentage of dominance states correctly predicted. To compare predictions of compositions (Table 1) we instead relied on probabilistic forecasting evaluation scores. Specifically, the energy score compares the composition $\mathbf{y} \in \mathbb{R}^2$ corresponding to the observation with the forecast distribution F defined by N samples $\mathbf{X}_1, \dots, \mathbf{X}_N, \mathbf{X}_i \in \mathbb{R}^2$. The formula for the energy score (denoted by ‘ES’ below) is

$$ES(F, \mathbf{y}) = \frac{1}{N} \sum_{i=1}^N \|\mathbf{X}_i - \mathbf{y}\| - \frac{1}{2N^2} \sum_{i=1}^N \sum_{j=1}^N \|\mathbf{X}_i - \mathbf{X}_j\|.$$

It is worth noting that this is a multivariate generalization of the more common continuous rank probability score, often used in epidemiology³⁹. Moreover, in the case of point forecast, it coincides with the mean absolute error. Thus, it can also be used for evaluating the M2 ‘H1–H3 alternation’ method, for which we do not have a confidence interval. The score is negatively oriented, so that it decreases when the forecast improves, and has a lower bound of zero for an ideal model. Furthermore, it is a strictly proper score, that is, designed to be optimal only when the forecast distribution coincides with the true distribution of the observations⁴⁰. Binary predictions—that is, dominance (yes/no) and negligibility (yes/no) of each subtype—were evaluated through the AUROC score (Table 2). It quantifies the overlap between the positive and negative classes and ranges from 0 to 1: 1 for an ideal model where the classes are perfectly separated, 0.5 for a random guess and 0 for the worst possible model that misclassifies all observations.

For the M5 ‘HVAR’ model, we tested lag = 1 and lag = 2. The results for the M5 ‘HVAR’ model in all tables are obtained using the lag with the lowest energy score for each group of countries, that is, lag = 1 for group II countries and lag = 2 for group I countries.

Other evaluation metrics were tested in the sensitivity analysis. They are described in Supplementary Table 1, while the results of these metrics are reported in Supplementary Tables 5 and 6.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The global flight network data are commercially available from the International Air Transport Association (IATA, <https://www.iata.org/en/contact-support>). All other data are publicly available. Data on positive influenza cases classified by type and subtype, as well as information on countries’ influenza seasons, were obtained via FluNet at <https://www.who.int/tools/flunet> (ref. 1). Meteorological data on temperature and relative humidity were obtained via the Climate Data store at <https://doi.org/10.24381/cds.6860a573>. Demographic data on national population sizes were retrieved from the United Nations at <https://population.un.org/wpp/>, and population data for urban agglomerations with more than 300,000 inhabitants and for capital cities were obtained from <https://population.un.org/wup/>.

Code availability

Analysis was implemented in R (version 4.3.2) and Python (version 3.8.5). Other than standard packages for data treatment, plots and calculations (mainly the Python packages pandas, matplotlib and numpy), we relied on the following packages for specific tasks: zComposition 1.4.0-1 (R) for zero imputation in the preprocessing of compositions⁷⁰; robCompositions 2.3.1 (R) for mapping compositions from the Simplex to the Euclidean space and back⁷¹; ternary (python) for drawing ternary plots⁷²; scipy 1.6.2 (python) for clustering analysis; sklearn 1.3.2 (python) for clustering analysis and computation of some of the prediction evaluation scores⁷³; R code developed by Lu et al.³⁸, on the basis of which we performed the VAR and HVAR predictions; scoringRules 1.0.2 (R) for calculation of proper scores for probabilistic forecast evaluation⁷⁴. Code and data for reproducible analyses are available via GitHub at <https://github.com/FrancescoBonacina/coupled-dynamics-flu-subtypes/>.

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Author contributions

F.B., P.-Y.B., V.C., O.L., M.T. and C.P. designed research; F.B. performed research; F.B., P.-Y.B. and C.P. analysed data; and F.B., P.-Y.B., V.C., O.L., M.T. and C.P. wrote the paper.

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Correspondence and requests for materials should be addressed to Chiara Poletto.

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¹Institut Pierre Louis d'Epidémiologie et de Santé Publique, Sorbonne Université, INSERM, Paris, France. ²Laboratoire de Probabilités, Statistique et Modélisation, Sorbonne Université, CNRS, Paris, France. ³CREST Laboratory, Groupe des Écoles Nationales d'Économie et Statistique, Ecole Polytechnique, Institut Polytechnique de Paris, CNRS, Palaiseau, France. ⁴Laboratoire de Sciences Actuarielle et Financière, Université Claude Bernard Lyon 1, Lyon, France. ⁵Department of Molecular Medicine, University of Padova, Padova, Italy. ⁶Present address: DONDENA Centre for Research on Social Dynamics and Public Policy, Bocconi University, Milan, Italy. ✉e-mail: chiara.poletto@unipd.it

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Cell population abundance	<i>Describe the abundance of the relevant cell populations within post-sort fractions, providing details on the purity of the samples and how it was determined.</i>
Gating strategy	<i>Describe the gating strategy used for all relevant experiments, specifying the preliminary FSC/SSC gates of the starting cell population, indicating where boundaries between "positive" and "negative" staining cell populations are defined.</i>

- ☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.

Magnetic resonance imaging

Experimental design

Design type	<i>Indicate task or resting state; event-related or block design.</i>
Design specifications	<i>Specify the number of blocks, trials or experimental units per session and/or subject, and specify the length of each trial or block (if trials are blocked) and interval between trials.</i>
Behavioral performance measures	<i>State number and/or type of variables recorded (e.g. correct button press, response time) and what statistics were used to establish that the subjects were performing the task as expected (e.g. mean, range, and/or standard deviation across subjects).</i>

Acquisition

Imaging type(s)	<i>Specify: functional, structural, diffusion, perfusion.</i>
Field strength	<i>Specify in Tesla</i>
Sequence & imaging parameters	<i>Specify the pulse sequence type (gradient echo, spin echo, etc.), imaging type (EPI, spiral, etc.), field of view, matrix size, slice thickness, orientation and TE/TR/flip angle.</i>
Area of acquisition	<i>State whether a whole brain scan was used OR define the area of acquisition, describing how the region was determined.</i>
Diffusion MRI	☐ Used ☐ Not used

Preprocessing

Preprocessing software	<i>Provide detail on software version and revision number and on specific parameters (model/functions, brain extraction, segmentation, smoothing kernel size, etc.).</i>
Normalization	<i>If data were normalized/standardized, describe the approach(es): specify linear or non-linear and define image types used for transformation OR indicate that data were not normalized and explain rationale for lack of normalization.</i>
Normalization template	<i>Describe the template used for normalization/transformation, specifying subject space or group standardized space (e.g. original Talairach, MNI305, ICBM152) OR indicate that the data were not normalized.</i>
Noise and artifact removal	<i>Describe your procedure(s) for artifact and structured noise removal, specifying motion parameters, tissue signals and physiological signals (heart rate, respiration).</i>

Volume censoring

Define your software and/or method and criteria for volume censoring, and state the extent of such censoring.

Statistical modeling & inference

Model type and settings

Specify type (mass univariate, multivariate, RSA, predictive, etc.) and describe essential details of the model at the first and second levels (e.g. fixed, random or mixed effects; drift or auto-correlation).

Effect(s) tested

Define precise effect in terms of the task or stimulus conditions instead of psychological concepts and indicate whether ANOVA or factorial designs were used.

Specify type of analysis: ☐ Whole brain ☐ ROI-based ☐ Both

Statistic type for inference

Specify voxel-wise or cluster-wise and report all relevant parameters for cluster-wise methods.

(See [Eklund et al. 2016](#))

Correction

Describe the type of correction and how it is obtained for multiple comparisons (e.g. FWE, FDR, permutation or Monte Carlo).

Models & analysis

n/a | Involved in the study

☐ ☐ Functional and/or effective connectivity

☐ ☐ Graph analysis

☐ ☐ Multivariate modeling or predictive analysis

Functional and/or effective connectivity

Report the measures of dependence used and the model details (e.g. Pearson correlation, partial correlation, mutual information).

Graph analysis

Report the dependent variable and connectivity measure, specifying weighted graph or binarized graph, subject- or group-level, and the global and/or node summaries used (e.g. clustering coefficient, efficiency, etc.).

Multivariate modeling and predictive analysis

Specify independent variables, features extraction and dimension reduction, model, training and evaluation metrics.